

Synthesis, microstructure and photocatalytic properties of {(1-x)BiOBr/(x)Bi₂₄O₃₁Br₁₀} composite for Ciprofloxacin photodegradation

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ABSTRACT:

In the general framework of photodegradation of pollutants in aqueous environments, a BiOBr (denoted as BOB) precursor was first obtained by hydrothermal synthesis, and then, thermally treated at 400, 450, 500, 550 and 600 °C. The decomposition of the BiOBr phase consisted in the progressive formation of Bi₂₄O₃₁Br₁₀ (Bi₂₄) phase leading to a BiOBr/Bi₂₄O₃₁Br₁₀ composite system. The phase transformation was analyzed using TDA-TGA, X-ray diffraction (XRD), Raman spectroscopy, scanning electron microscopy (SEM) coupled with energy dispersive X-Ray analysis (SEM-EDX). The optical properties were investigated with UV-visible diffuse reflection spectroscopy (DRS). The band and electronic structures of the single phases BiOBr and Bi₂₄O₃₁Br₁₀ were determined from calculations using Density Functional theory (DFT). These BiOBr/Bi₂₄O₃₁Br₁₀ composite systems were used to photodegrade the antibiotic ciprofloxacin (CIP) under simulated-sunlight irradiation. The optimal photocatalytic efficiency was obtained from the BOB-450 mix system obtained at 450 °C, characterized by approximately 22 mol % of Bi₂₄ phase. The variable photocatalytic performance was ascribed to the flower-like porous morphologies and the probable existence of homo and heterojunctions in this system. The main parameters influencing the photocatalytic activity of BiOBr/Bi₂₄O₃₁Br₁₀ composites such as pH, CIP content, photocatalyst amount and irradiation type were analyzed. The photocatalytic mechanism and the factors driving the enhanced performance of BOB-450 system was proposed.

Keywords: Hydrothermal synthesis; BiOBr/Bi₂₄O₃₁Br₁₀ composite; DFT calculations; photocatalysis; ciprofloxacin photodegradation

BIOGRAPHY:

I am a PhD candidate in cotutelle between the University of Toulon, France, and Ibn Zohr University, Morocco, in the field of materials science. My research focuses on the synthesis, characterization, and optimization of bismuth-based oxide and oxyhalide heterostructures for the photocatalytic degradation of organic pollutants. I have experience in thermal, structural, morphological, and optical characterization techniques, including XRD, SEM, and spectroscopy, as well as DFT-based analysis of material properties. I am also involved in higher education teaching and have contributed to several international publications and conferences related to functional materials and environmental applications.

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